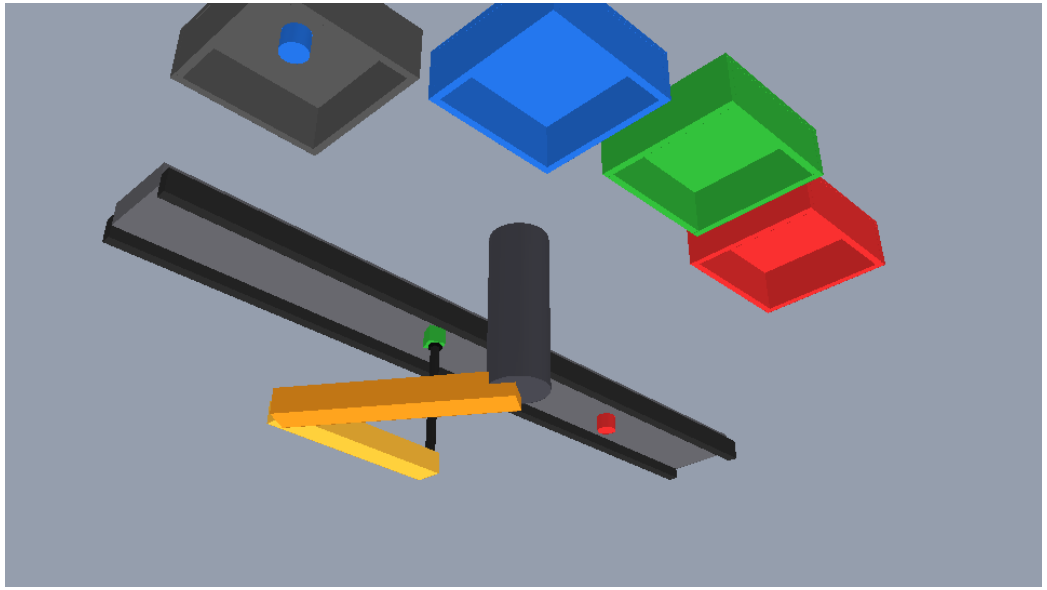
Vision-Guided Sorting Arm on a Conveyor

Predictive visual servoing of a SCARA arm: colour and shape sorting in CoppeliaSim

Project P14 | Group G10 | Batch B4 | CSAI4014 Algorithms for Intelligent Systems and Robotics

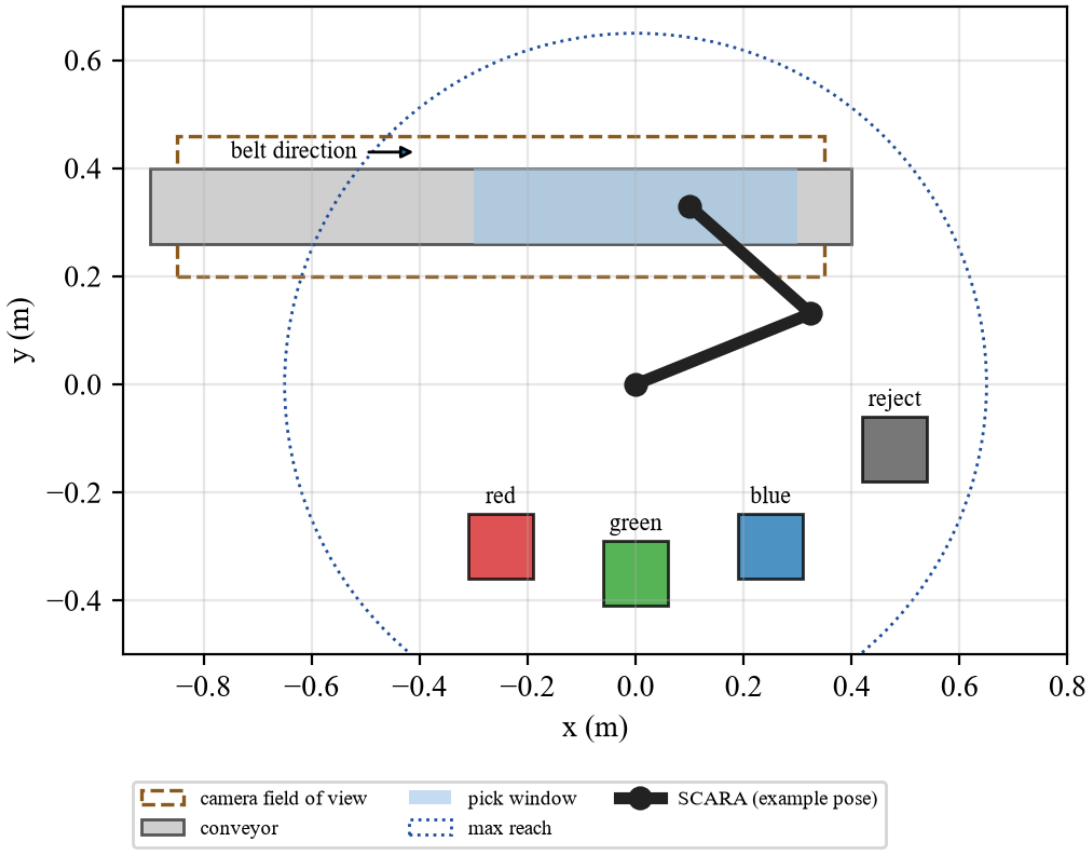
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Submitted to: Course Faculty, CSAI4014, School of Computer Science, UPES, Dehradun | October 2026

Problem and Motivation

- **Task:** detect parts by colour AND shape on a moving belt; cubes to R / G / B bins, cylinders (wrong shape) to a reject bin
- **Domain:** quality inspection and sorting in manufacturing
- **Hard part:** the part moves during the 80 ms camera delay and the arm motion; belt speed ripples
- **Today:** many cells stop the belt and waste throughput



Objectives (measurable)

#	Objective	Target	Achieved
O1	Classify colour + shape	>= 99 % at noise <= 25	100 % / 100 %
O2	Grasp error, moving belt	<= 5 mm to 0.35 m/s	2.7 to 3.4 mm
O3	Sorting success, robust to ripple	>= 95 % at <= 0.20 m/s	100.0 %; 15 % ripple 88 %
O4	Throughput vs stop-and-go	>= 2 x at 0.25 m/s	2.2 x
O5	Inspection: wrong accept / reject	<= 0.1 %	1 of 3213 cylinders, 0 rejects

Assumptions: rendered orthographic camera, suction = attach constraint, kinematic conveyor, 3 colours, 2 shapes, simulation only.

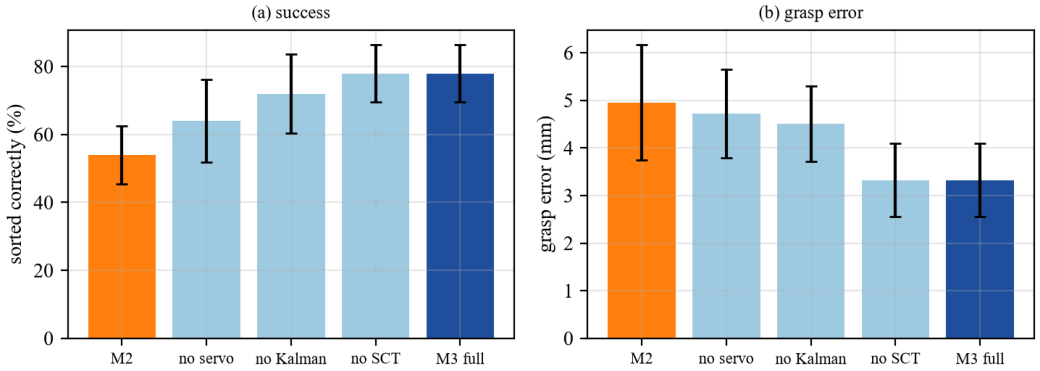
Existing Solutions and the Gap

Approach	Strength	Limitation
Stop-and-go conveyor	accurate, simple	about half the throughput
Encoder line tracking (FANUC, ABB)	fast, mature	closed software
Visual servoing theory	feedback cancels model error	rarely applied to delayed moving parts
Kalman tracking	optimal estimate	grasp benefit rarely quantified

Gap: no open study that combines tracker + intercept planner + visual servo and compares it with baselines on two physics engines.

Novelty

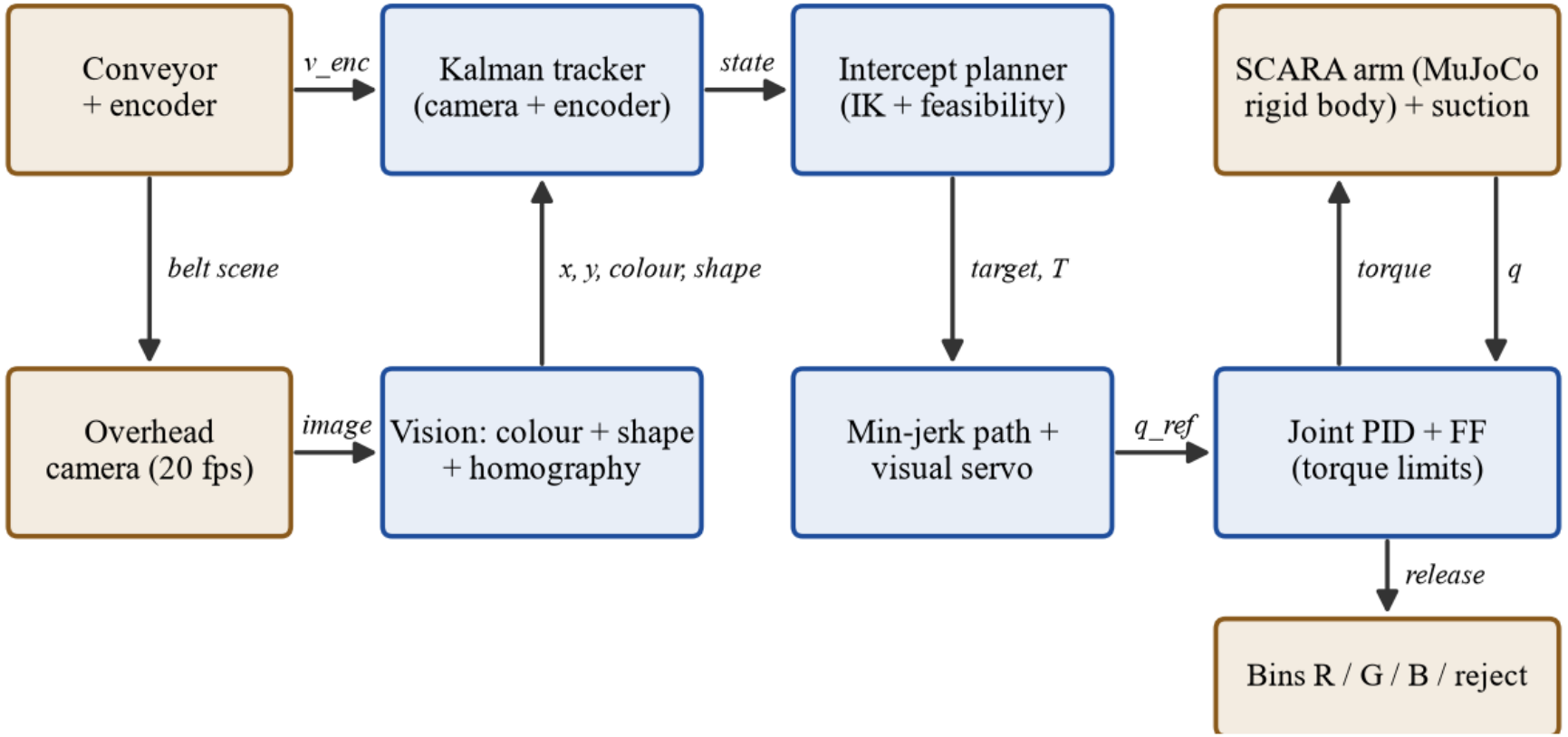
- **N1** Visual-servo descent: grasp error 4.7 to 3.3 mm, success 64 to 78 %
- **N2** Kalman + encoder: success 70 to 78 % vs line fit ($p < 0.001$)
- **N3** Robust to 15 % ripple: 88 % vs 49 %
- **N4** Same code on CoppeliaSim and MuJoCo, 890 + runs
- **N5** Negative result: scheduling rule gives no gain



Syllabus Mapping

AISR topic (Project List)	Where it is used
Robotic Perception	vision sensor, HSV colour, contour shape, homography, Kalman tracking
D-H Transformations	D-H table, matrices A1 A2 A3, FK / IK, Jacobian
Closed-Loop Control	joint PID + feedforward, visual-servo descent
Application Domains	quality inspection and sorting in manufacturing

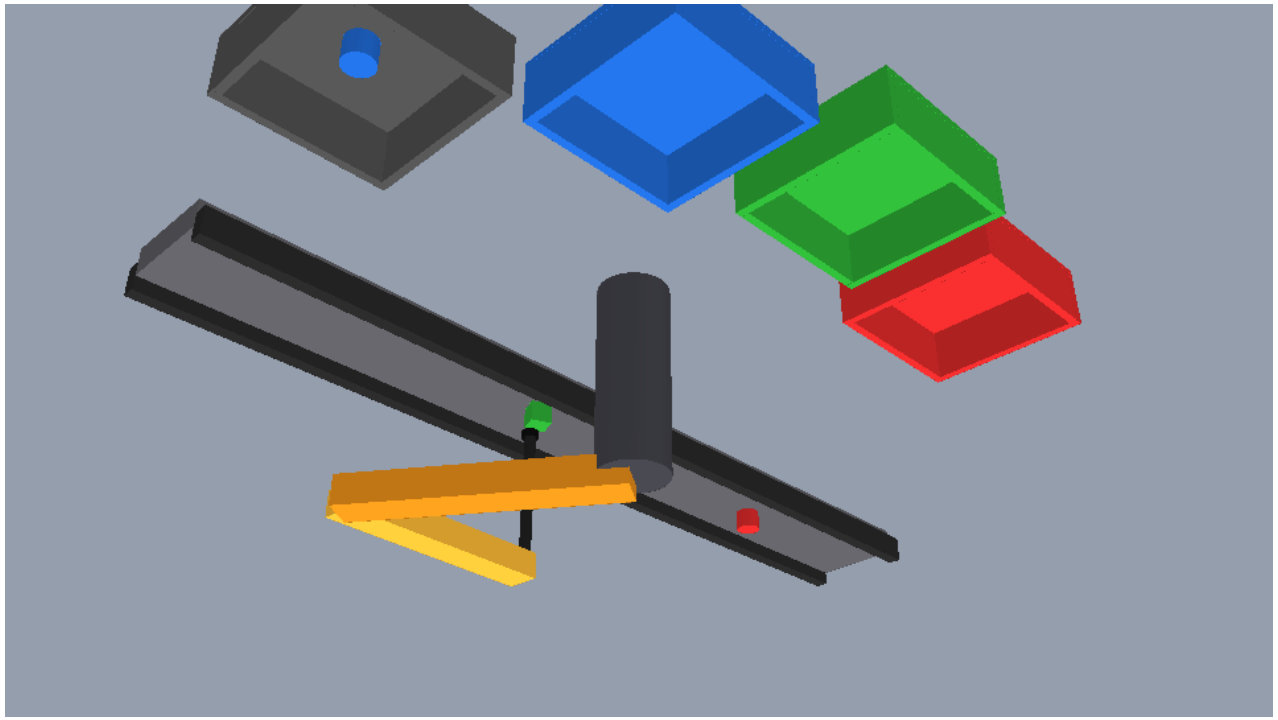
System Architecture



20 fps vision sensor, 80 ms latency, encoder fusion, joint control inside the physics step

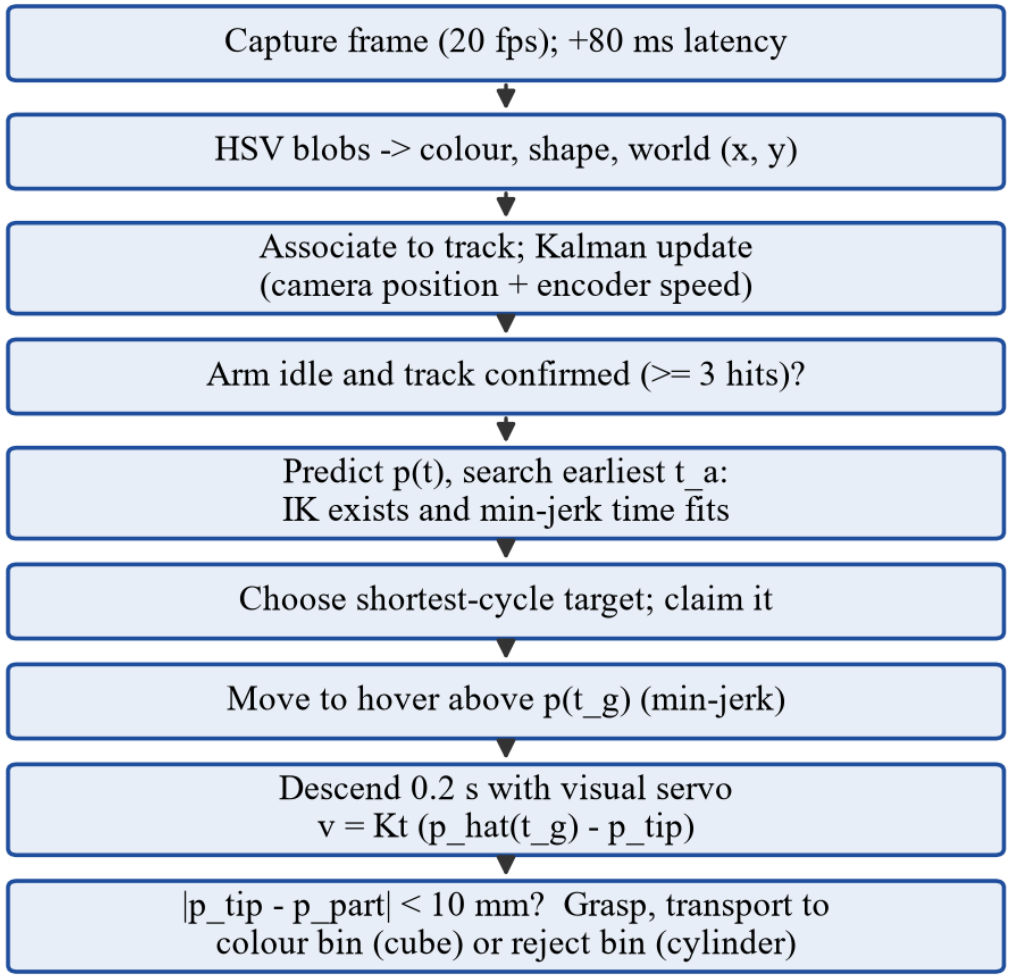
Robot and Environment Model

- **Arm:** SCARA, L1 0.35 m, L2 0.30 m, suction tool
- **D-H:** table + matrices, closed-form IK
- **Physics:** CoppeliaSim (Bullet), torque limits 45 / 18 N m
- **Belt:** 0.05 to 0.35 m/s, 3 % ripple
- **Parts:** 75 % cubes, 25 % cylinders



CoppeliaSim scene during the descent

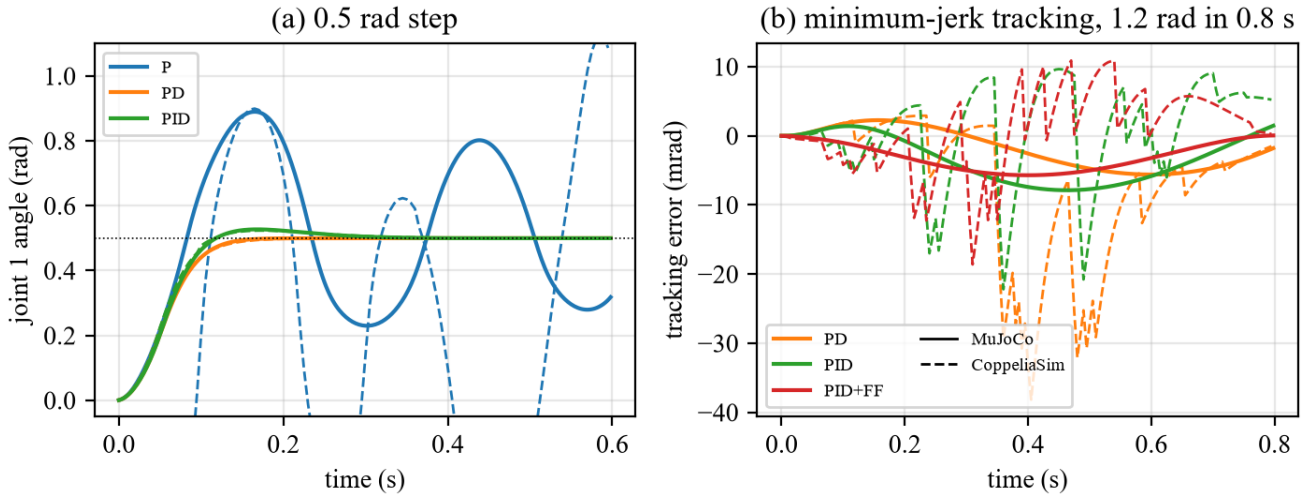
Algorithm 1: Intercept Planning



- **Predict:** p(t) = p_hat + v_hat (t - t_f), compensates 80 ms
- **Search:** earliest t_a with IK solution and min-jerk time T <= available
- **Select:** shortest cycle among feasible targets
- **Skip:** parts that cannot be reached in time

Algorithm 2 and Controller Design

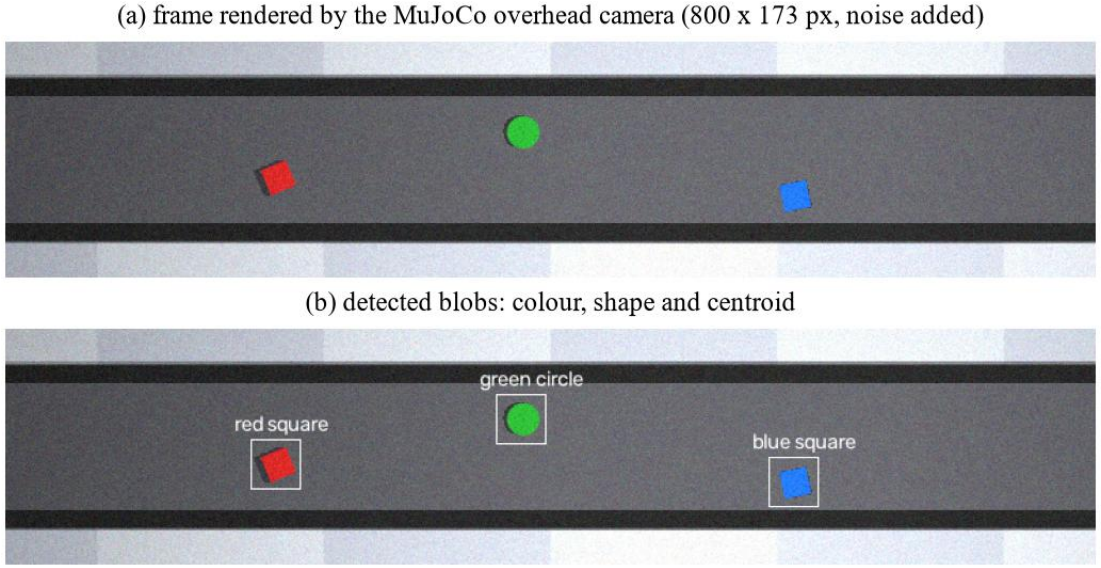
- **Servo:** v_cmd = Kt (p_hat(t_g) - p_tip), dq = J^-1 v_cmd
- **PID:** wn = 50 rad/s, zeta = 1, Ki = 0.2 Kp wn
- **FF:** J_n q_r'' + B_n q_r' cuts the PID tracking error by 28 %
- **In CoppeliaSim:** PID + FF runs in a joint-callback script



Max tracking error (MuJoCo arm): PID 7.9 mrad, PID + FF 5.7 mrad

Simulation Setup

- **Platform:** CoppeliaSim V4.10 (assigned) + Python + OpenCV
- **Second engine:** same code on MuJoCo for the large statistics
- **Run:** python demo_cs.py
- **Check:** success = part physically in its bin

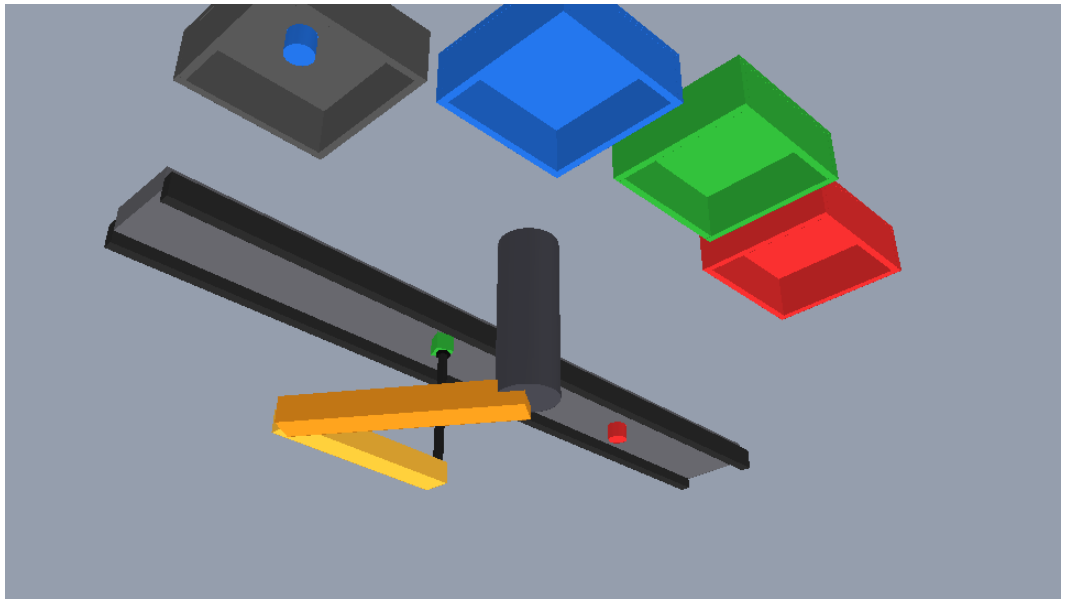


Vision-sensor frame and detections

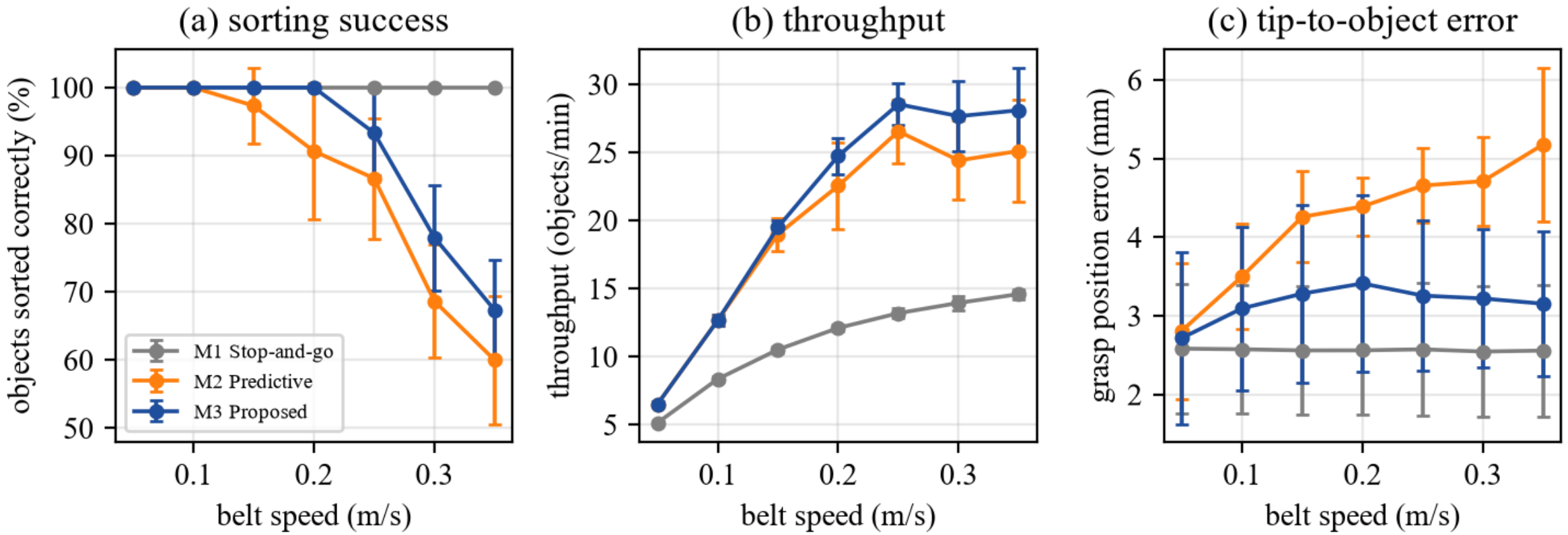
Live Demonstration

Switch to CoppeliaSim now

- **Run:** python demo_cs.py
- **Show:** arm waits ahead, descends while tracking; cubes to colour bins, cylinders to reject bin
- **Compare:** --method m2 --speed 0.3 shows misses



Results 1: Speed Study



M3: 2.7 to 3.4 mm grasp error, 2.2 x the throughput of stop-and-go at 0.25 m/s. Above 0.25 m/s one arm cannot serve every part.

Results 2: Comparison of Methods

Method (0.25 m/s)	MuJoCo success %	CoppeliaSim success %	Grasp error mm
M1 stop-and-go	100	100	2.6
M2 predictive	87	85	4.7
M3 proposed	93	95	3.3

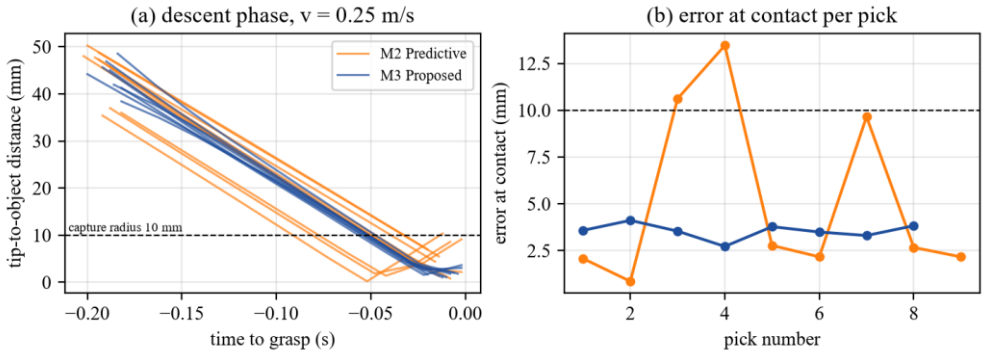
- **Ripple 15 %:** M3 88 % vs M2 49 % (30 seeds: $p < 0.001$)
- **Ablation:** M2 55, no servo 67, no Kalman 70, full 78 %
- **Both engines:** same ranking M3 > M2

Team Contribution

Member	Technical module
Aditya Khatri	SCARA kinematics, D-H, CoppeliaSim and MuJoCo cells, visual servo
Simran Jeet Gill	Vision sensor image, HSV detector, shape classifier, homography
Divya Israni	Kalman tracker, intercept planner, ablation
Amritanshi Singh	PID + feedforward, joint callback, tuning, step tests
Anchit Saxena	Experiments, stop-and-go baseline, renders, live demo

Conclusion and Industry Use

- **Findings:** visual servo and Kalman both help; scheduling did not
- **Domain:** quality inspection and sorting in manufacturing
- **Real robot:** triggered camera, hand-eye calibration, vacuum delay
- **Future:** second arm, learned detector, ROS and real-arm test
- **Team:** we taught each other our modules



Same trial: M2 has misses above 10 mm, M3 none

Thank You

Questions?

GitHub: https://github.com/Aditya-Khatri-7/Robotics-assignment-project-b4

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